

Evolution by natural selection Student No. 530

Introduction

Although the Darwinian form of evolution has been an inescapable concept in biology and natural selection is widely used in different fields (Michael, 2016), there are still a great many people still in disagreement with the evolutionary theory. The lack of acceptance of the evolutionary theory may limit students' ability to talk about socio-scientific issues based on biology. (Fowler & Zeidler, 2016) Besides, Steve Jones, a geneticist at University College London, said that human evolution by natural selection has weakened. (Bootle, 2011) Thus, an inquiry into whether this fundamental concept of natural selection still works is worthwhile to be done. From my perspective, people are still evolving by natural selection. To reach this conclusion, I will first explain why humans are still evolving and how natural selection differs from other mechanisms of evolution. Then, I will give evidence in support of the ideas that humans are still evolving by natural selection and critically analyze the ideas. Lastly, we can draw a conclusion based on all the evidence and our critical thinking.

Evolution as fact and theory

We can divide our topic into two questions:

1. Are humans are evolving?
2. If humans are still evolving, is the evolution caused by natural selection?

Actually, these two questions reflect the two related but different sort of evolution: evolution as a pattern and evolution as a process, as is described in *Biology: A Global Approach*. In the book, evolution is defined as descent with modification, or more narrowly the genomic change from generation to generation, which is a pattern revealed by observations, while evolution as a mechanism is the theories about how the pattern is formed. For a fact, we need to check the observations, while for a theory, we need to check whether the hypotheses can fit the facts well. (Campbell, et al., 2016)

As to evolution as a pattern, surely humans are still evolving. With the help of the huge amount of gene data, shows that certain changes in human genomes. For example, a study conducted by John Hawks shows the ongoing adaptive evolution of human with an acceleration of positively selected alleles. (Hawks, et al., 2007). In some human populations, lactase persistence is found to be more common in some populations. (Bersaglieri, et al., 2004) All of these

indicate a certain degree of evolution in humankind.

As for the mechanisms of evolution, today, other basic mechanisms are also used to cause the evolutionary changes, such as genetic drift, migration, inbreeding, and mutation. (Casillas & Barbadilla, 2017) To distinguish natural selection from other possible mechanisms, we need to find out the differences among them. The first difference is that natural selection has certain directions, vis-à-vis mutation and genetic drift which are random. (William, 1974) Most importantly, natural selection is believed to be the only mechanism that continuously leads to adaptation. (Campbell, et al., 2016)

However, adaptation is not necessarily caused by natural selection. If there are no quantitative relationships amongst sampling errors, selection pressure, and enough chance of random mutation, natural selection will not be effective. (William, 1974) Other mechanisms may cause the similar results. For example, the imprinting effect and matching habitat choice are considered to be able to cause that grasshoppers living in different parts of a sidewalk are in the color fitting the background color. (Edelaar, et al., 2015) However, this kind of adaptation is not the what natural selection tries to explain, because if there is imprinting effect or matching habit choice, there will be no selective pressure. Also, the group is too small in size to cause a long-term evolution.

Besides, today we understand natural selection differently from Darwin, or else we would not use the terms like genotype or phenotype. We will mainly talk about how we understand natural selection in this essay. Thus, natural selection is a process causing organism's adaptation to the environment by selectively accumulating genetic and phenotypic changes." (Encyclopædia Britannica Editors, 2018) Then, this definition gives rise to many testable hypotheses. Although not every hypothesis is tested, I will try to find a generalized but reasonable conclusion.

Localized Natural Selection

According to the discussion above, adaption can happen as a result of natural selection. And because the fitness in natural selection does not necessarily mean the 'good' traits such as being strong and being swift; therefore, what we usually think is a 'bad' phenotype or genotype will be favored by the environment. People with the traits are more likely to overcome the struggle for life, and thus gain reproductive success. Such selection should be localized in a certain environment different from most other environments and continuously make people in the area evolve.

One of the most famous examples of the kind of selection is on the gene that causes sickle cell disease. People with the hereditary sickle cell disease were found to be less likely to be infected by malaria in 1952, (Serjeant, 2010) In the regions where malaria is hyper-endemic, the sickle-cell trait was more common than other areas without the disease. (Allison & D. Phil., 1954) Researchers from Dartmouth College have used a mathematical model and recent years' data and still found that the heterozygous carrier of the diseased genes bears less selection pressure. (Liddell, et al., 2014)

However, although the higher frequency of the sickle cell trait can also be due to the founder effect, which means the higher frequency of certain genes in a population is due to the original small group with some members who carry the genes. Yet, another study in African population indicates natural selection driven by other diseases, such as kidney diseases, has brought about changes in genomes, (Gomez, et al., 2014) which lowers the possibility because people with many diseased traits are less likely to survive long.

Today, many areas are still threatened by malaria. Although the deaths caused by malaria in Africa has seen a rapid decrease in recent years, for specific countries like Chad and Central Africa the deaths caused by malaria have not significantly decreased and the 10-year accumulative mortality in 1980-2010 remains above 20%, despite the rapid decrease of death rates caused by malaria in Africa and globally. (Murray, et al., 2012) Thus, we cannot deny that in the areas with epidemics of diseases, malaria is still causing natural selection. Another similar case is a protective prion protein variant found in Papua New Genuine in 2009 is considered as a new sign of localized natural selection due to kuru, an epidemic prion disease. (Simon Mead, et al., 2009) As we can believe, such localized natural selection will still continue.

The Inheritable Behavior

For those complicated human preferences, genes are still considered as an important reason. Research on twins in Sweden and the United States finds that people's preference for risk is broadly inheritable. (Cesarini, et al., 2009) Also, genetics is shown to influence political attitudes and ideologies in some twins in Australia, although the researchers also admit that genetic influence on party identification is lesser. (Alford, et al., 2005) Some researchers have brought up a model in which how positively correlated genotypes cause positive selection driving people's evolution, making people more likely to form social ties. (Christakis & Fowler, 2014)

Although the selection based on inheritable preferences is made by humans, it

including sexual selection should still be considered as a kind of natural selection, or else we will face the questions like why some animals are predators while some are not. If some animals are predators because of genes, certain human behavior is also found to be related to certain genes. If it is because of the influence of the group, social selection including sexual selection can also be a result of that kind of influence. According to the definition of natural selection which is given previously, natural selection is more about reproductive success. Thus, sexual selection as well as some other social selection should not be excluded from natural selection by us.

Empirically sexual selection as a part of social interactions should be still affecting the evolution of humankind, yet we need more academic report to confirm it. However, for other types of human behavior, to know whether they are inheritable requires further research. And how certain social selection participates in the evolution needs more investigation. There are results of social selection such as altruism and so on, yet, more research should be done before we can reach a conclusion on whether such selection is still ongoing.

Conclusion

As a matter of fact, humans are still evolving. And there is also no denying in the areas with high prevalence of certain diseases, natural selection driven by the diseases is still making the population in the areas evolve. Besides, social selection, including sexual selection, as I believe, should be part of natural selection, yet further discovery is needed to understand their role in human evolution and to find out whether the selection is still ongoing.

Also, some research indicates that although the natural selection is still influencing the evolution of humankind, the intensity of selection in the surveyed population has become lower, despite the direction still not changed, based on data from the certain female population in America since 1948. But the surveyed population are mostly of European ancestry and consist of women only. (Byars, et al., 2010) And I believe that we still need further research on a population with more diversity to reach a solid conclusion for whether natural selection in humankind has slowed down.

Bibliography

- Alford, J. R., Funk, C. L. & Hibbing, J. R., 2005. Are Political Orientations Genetically Transmitted?. *The American Political Science Review*, May, 99(2), pp. 153-167.
- Allison, A. & D. Phil., B., 1954. The distribution of the sickle-cell trait in East Africa and elsewhere, and its apparent relationship to the incidence of subtertian malaria. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, 48(4), pp. 312-318.
- Bersaglieri, T. et al., 2004. Genetic Signatures of Strong Recent Positive Selection at the Lactase Gene. *The American Journal of Human Genetics*, 74(6), pp. 1111-1120.
- Bootle, O., 2011. *Are humans still evolving by Darwin's natural selection?*. [Online]
Available at: <http://www.bbc.com/news/science-environment-12535647>
[Accessed 28 Jan. 2018].
- Byars, S. G., Ewbank, D., Govindaraju, D. R. & Stearns, S. C., 2010. Natural selection in a contemporary human population. *Proceedings of the National Academy of Sciences*, 107(suppl 1), pp. 1787-1792.
- Campbell, N. A. et al., 2016. Chapter 22 Descent with modification: A Darwinian View of Life. In: *Biology: A Global Approach*. s.l.:Pearson Education Ltm., pp. 462-469.
- Casillas, S. & Barbadilla, A., 2017. Molecular Population Genetics. *Genetics*, 1 Mar., 205(3), pp. 1003-1035.
- Cesarini, D. et al., 2009. Taking, Genetic Variation in Preferences for Giving and Risk. *The Quarterly Journal of Economics*, May, 124(2), pp. 809-842.
- Christakis, N. A. & Fowler, J. H., 2014. Friendship and natural selection. *Proceedings of the National Academy of Sciences of the United States of America*, 111(10796), p. 10901.
- Edelaar, P., Baños-Villalba, A., Pablo-Quevedo, D. & Escudero, G., 2015. *Empirical evidence that natural selection is not the only mechanism of adaptive evolution*. Lausanne, Switzerland, 15th Meeting of The European Society for Evolutionary Biology.
- Encyclopædia Britannica Editors, 2018. *Natural selection*. [Online]
Available at: <https://www.britannica.com/science/natural-selection>
[Accessed 31 Mar 2018].
- Fowler, S. R. & Zeidler, D. L., 2016. Lack of Evolution Acceptance Inhibits Students' Negotiation of Biology-based Socioscientific Issues. *Journal of*

Biological Education, 21 Mar..50(4).

Gomez, F., Hirbo, J. & Tishkoff, S. A., 2014. Genetic Variation and Adaptation in Africa: Implications for Human Evolution and Disease. *Cold Spring Harbor Perspectives in Biology*, 6(7), pp. a008524-a008524.

Hawks, J. et al., 2007. Recent acceleration of human adaptive evolution. *Proceedings of the National Academy of Sciences of the United States of America*, 104(52), pp. 20753-20758.

Liddell, C., Owusu-Brackett, N. & Wallace, D., 2014. A Mathematical Model of Sickle Cell Genome Frequency in Response to Selective Pressure from Malaria. *Bulletin of Mathematical Biology*, 76(9), pp. 2292-2305.

Michael, M., 2016. *Evolution by Natural Selection: Confidence, Evidence and the Gap*. 20151007 ed. Boca Raton, FL: CRC Press, Taylor & Francis Group.

Murray, C. J. L. et al., 2012. Global malaria mortality between 1980 and 2010: a systematic analysis. 4-10 Feb., 379(9814), pp. 413-31.

Reznick, D. N., 2010. *The Origin then and now: an interpretive guide to the Origin of species*. Princeton(NJ): Princeton University Press.

Serjeant, G. R., 2010. One hundred years of sickle cell disease. *British Journal of Haematology*, 11th Nov., 151(5), pp. 1365-2141.

Simon Mead, M. et al., 2009. A Novel Protective Prion Protein Variant that Colocalizes with Kuru Exposure. *The New England Journal of Medicine*, Volume 361, pp. 2056-65.

William, G. C., 1974. *Adaptation and Natural Selection: A Critique of Some Current Evolutionary Thought*. 1996 ed. Princeton, New Jersey: Princeton University Press.